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SUBJECT OF THE INVENTION

[**0001**] The present invention relates to a radiation measuring device, relates to a device for measuring in particular separating the radon-Tron.

[**0002**] Background of the Invention is a radioactive isotope radon (222Rn: atomic number 86) and Tron (220Tn: atomic number 86) is present as a gas in the air. For this reason it is necessary to pay attention to the body exposure by inhalation.

[**0003**] In recent years, radon-Tron as environmental radiation that is emitted from the concrete wall has been a problem, radiation handling facilities there has been a growing need for measurement of radon-Tron also in well general of the building of the room within or basement, etc. .

[**0004**] In FIG. 5, radon decay chain (A) and Tron of decay series (B) are shown. Only the left edge of the radon and thoron is a gas, and their daughter is a solid. As will be described later, their daughters excluding radon and thoron are positively charged by emitting α rays.

[**0005**] Figure 4 is an example of a conventional radon thoron measuring device is shown. In the figure, the sample gas (measurement gas) is first sent to a dust filter 10 is a large foreign matters such as dust are removed. Next, the ion pre-recipe stator 12, ions contained in the sample gas by using the electric field is removed. This, in the next ion chamber 14, are unwanted ions provided to prevent from being counted. As a result, it has become a result of daughter nuclides of radon daughter nuclides and Tron is also removed.

[**0006**] In the ionization chamber 14, the α rays radon and thoron are released upon disintegration, the gas in the ion chamber 14 is ionized, the center electrode 14a by the electric field at this time resulting charge is provided in the ionization chamber It is the current collector in. Thus the output signal from the ion chamber 14 occurs through the vibrating capacitance electrometer head 16, the vibration is transmitted to the capacitor electrometer 18, which is the concentration in conjunction with radon thoron displayed by. Incidentally, the flow meter 20 and the pump 22 are connected to the ion chamber 14. There is a such as those that the integrated measurement of the order of January to six months using, for example, other as a TLD.

[**0007**] [SUMMARY OF THE INVENTION However, in the above conventional apparatus, could hardly be measured radon and thoron separately. Further, in the above conventional apparatus, since radon daughters and thoron daughters are removed by ion pre recipe theta 12, it could not be measured including those nuclides. Incidentally, it is desirable to improve the measurement sensitivity of the radon-Tron.

[**0008**] The present invention has been made in view of the above conventional problems, and an object thereof is respectively separately measure radon and thoron.

[**0009**] Further, the present invention provides real-time radon daughters and thoron daughters, and to realize simple, economical measurement.

[**0010**] Furthermore, the present invention aims to improve the measurement sensitivity.

[**0011**] To achieve Disclosed solving means for the above object, the invention according to claim 1, and at least one radiation detection unit, a signal processing unit for treating a detection signal from the radiation detecting unit, the a radiation measuring apparatus comprising, the radiation detection unit, detector and, with one side electrode disposed in proximity to said radiation incident window, the one side electrode being spaced from said radiation incident window having a radiation entrance window wherein the facing is arranged and the other side electrode, with a polarity to collect charging nuclide to said one side electrode, and means for applying a voltage between said one side electrode and the other side electrode on said one side by introducing a measurement gas between the electrode and the other side electrode and performing detection of radiation by said detector.

[**0012**] According to a second aspect of the invention, a first detection unit for detecting the radon, and a second detection unit for detecting the radon and its daughters and thoron and their daughter, from the first and second detection units a radon thoron measuring apparatus including a signal processing unit for processing the output detection signal, the first detection unit has a first and ionization chamber having an α-ray incident window, the α-ray incident window a first gas introducing chamber formed next to the first ion chamber across the disposed gas inlet of the first gas introducing chamber, and a filter for transmitting radon gas only, the first gas first a first of the one-side electrode disposed in proximity to said radiation incident window in the introduction chamber, which is disposed opposite to the one side electrode in the first gas introducing chamber being separated from said radiation incident window 1 and the other side electrode, with a polarity to collect the radon daughter nuclides generated by the decay after filter transmission in the one side electrode, a first voltage for applying a voltage between said one side electrode and the other side electrode wherein the applying means, and said second detection unit, a second ionization chamber having an α-ray incident window is formed next to the second ionization chamber

across said α-ray incident window, a gas inlet from a second gas introducing chamber to the measurement gas is introduced, the second one side electrode disposed adjacent to the second of said radiation incident window in the gas introduction chamber, the second gas introducing chamber a second other-side electrode disposed to face the second one side electrode being spaced from said α-ray incident window in, collecting the radon daughters and thoron daughters in the second one side electrode with polarity, characterized in that it comprises a and a second voltage applying means for applying a voltage between the said and the second one side electrode and the second on the other side electrode.

[**0013**] According to a third aspect of the invention, the signal processing unit is characterized by comprising means for determining radon count, means for determining Tron counts, the.

[**0014**] [Action] According to the arrangement according to claim 1, the voltage between the arranged spaced from the radiation entrance window to be positioned proximate the one side electrode and the radiation entrance window of the detector other side electrode It applied. Thereby, an electric field between the electrodes is formed. Voltage application, whereas the side electrode is performed such that a negative electrode, charged radionuclide will be gathered to one side electrode. The radiation from radionuclides collected via a radiation entrance window and enters the detector, whereby the radionuclide is detected.

[**0015**] Since the form an electric field to collect the charged species to the detector vicinity, it is possible to increase the detection efficiency of the species, i.e. enhanced detection sensitivity. Incidentally, radionuclides uncharged via a radiation entrance window are observed as long as the radiation is detected, whereas, including radionuclides and other ions not radioactive because they do not emit radiation, can eliminate the influence of them. In this way it is possible to detect with the parent nuclide to for example the daughter nuclides of radon and thoron with a high sensitivity to. It should be noted that the detection efficiency of the daughter nuclide to the parent nuclide has become known by an experiment or calculation or the like. In addition, the ratio of the parent nuclide and daughter nuclides in equilibrium is constant.

[**0016**] According to the arrangement according to claim 2, the first detection unit is roughly composed of a first ionization chamber and the first gas introducing chamber, only radon in the measurement gas is detected. That is, filter arranged in the gas inlet of the first gas introducing chamber, since only the radon (radon gas) to be measured is intended to transmit, to the first gas introducing chamber, radon and the passed through the filter and only the radon daughter nuclides produced by nuclear decay from radon is present.

[**0017**] Then, is disposed first on one side electrode proximate the first α-ray incident window of the ionization chamber, at the same voltage are applied between the first other-side electrode, the

electric field between the electrodes There has been formed. The first one side electrode is the negative electrode, first the other side electrode is a positive electrode.

[**0018**] Thus, in the first detection unit, alpha rays emitted from radon introduced into the first gas introducing chamber is detected by the first ionization chamber via the alpha ray incident window. On the other hand, radon progeny newly generated in the gas introduction chamber, the formed electric field is captured in the first one-side electrode, where the resulting α-rays in the first ionization chamber via the α-ray incident window It is detected on.

[**0019**] Here, by measuring the radon daughter nuclides generated by the gas introduction chamber, since nothing but to measure the radon before entering the gas introduction chamber, as a result, the detection sensitivity of radon in a measurement gas It can be improved to become.

[**0020**] As the radon extraction filter, filter paper or the like having a predetermined thickness is used. Its thickness is set to for example, a gas passing time of is about several minutes to 10 minutes. According to this structure, with a half-life 55 seconds Tron, almost all thoron was allowed to disintegrate in pass filter 216Po (atomic number: 84) is solid to alter (the 216Po be 0.15 seconds half-life in the 212Pb (atomic number: 82) of the change in), can be captured by the filter by such a sol. In other words, all of the species of Tron series, including the Tron is blocked, as well as this, is arrested for daughter nuclides of radon is also solid.

[**0021**] The second detection unit, except filters for radon extraction has a first detection unit and basically the same structure, that the second gas introducing chamber and radon and radon daughters all of Tron and Tron daughter nuclide is introduced. Thus, by the action of the electric field, the second of the one-side electrode radon progeny and thoron daughter nuclide is charged nuclides are captured. Thus, alpha rays from their daughter nuclides are detected with high sensitivity at a second ionization chamber via the alpha ray incident window, at the same time, also alpha rays from radon and thoron present in the second gas introducing chamber It is detected by the second ionization chamber through the α-ray incident window. Therefore, it is possible to obtain total detection result by the second ionization chamber.

[**0022**] In either unit, electric field is formed by a pair of electrodes, it is possible to collect the daughter nuclides, it is possible to increase the detection sensitivity.

[**0023**] According to the configuration of claim 3 wherein the count value of the first detection unit and counting value of radon from the detection signal of the second detection unit and thoron are determined. That is, it is possible to realize a separate measurement of radon and thoron. It should be noted, is the first daughter of Tron 212Pb (atomic number: 82) is a half-life of 0.15 seconds, since the collapse immediately after Tron collapse, so that the detection pulse is two consecutive.

Using this, a predetermined time (e.g., 1 second) by counting the number of pairs of two pulses within are continuous, as a result, it becomes possible to measure the concentration of thoron.

[**0024**] Example will be described below on the basis of a preferred embodiment in the drawings of the present invention.

[**0025**] FIG. 1 shows the overall configuration of a radiation measuring device according to the present invention. Radiation measuring apparatus of this embodiment is the radon thoron measurement device.

[**0026**] 1, the radiation measuring device is roughly comprised of a detector unit 24 and the signal processing unit 30. Here, the detection unit 24 is constituted by the first detection unit 26 and the second detection unit 28, its details are shown as a sectional view in FIG.

[**0027**] 2, the first detection unit 26 serves to detect only radon in a measurement gas, the second detection unit 28, detects the radon and its daughters and thoron and their daughter nuclides in the measurement gas it is intended to (see Figure 5). The first detection unit 26 and the second detection unit 28 is identical in basic structure to each other, that the first detection unit 26 has a radon extraction filter 32 is different.

[**0028**] In the first detection unit 26, the bottom plate 34a of the metal detector casing 34 having a cylindrical shape, a circular opening 34b of the gas inlet are formed. Each opening 34b is, as described above, are closed by radon extraction filter 32 for transmitting only radon. As the radon extraction filter 32, for example, filter paper having a thickness that transmits radon gas in about 1 to 10 minutes are employed. That is, while the half-life of radon is 3-4 days, since the half-life of thoron is 55 seconds, by using a transformation from the difference and the gas of the half-life to the solid, the nuclear species other than radon it is intended to prevent. Only radon and thoron is a gas, because their daughter is a solid, Tron stops moving at a sol shape in the filter paper if collapse. Of course, the collapse is carried out according to a probability, radon is also collapse in the stochastic, because the number is small enough not to be compared, it can be virtually ignored.

[**0029**] Thus, by radon extraction filter 32, radon progeny, thoron and its daughter nuclide is blocked, only radon as a gas is to be entered in the detector case 34.

[**0030**] Within the detector casing 34, the ionization chamber 36 is disposed above the figure, also, the gas introducing chamber 38 is formed below in FIG. The gas introduction chamber 38, which may be formed as a gas-tight space, in this embodiment the internal gas is adapted to be evacuated naturally.

[**0031**] α-ray incident window 36A of the ionization chamber 36 consists of a thin material that transmits α rays, for example, a like mica. On the inside surface 40 of the α-ray incident window 36A, the electrically conductive coating is applied, at the same ionization chamber container is constituted by a shell 42 of your Wangata. Conduction is achieved with conductive Koteiingu and shell 42.

[**0032**] Within the shell 42, while being insulated from the shell 42 is ring-shaped collector electrode 44 is disposed. The ionization chamber 36 is filled with air or an inert gas substantially one atmosphere, the gas is ionized by the incident of the α rays, electrons generated at that time by the action of an electric field, is the current collecting the collector electrode 44. Therefore, the high voltage between the shell 42 and the collecting electrode 44, for example, 500V (50~1500V is used) is applied.

[**0033**] The inner surface of the detector casing 34 is disposed an insulating member 46, it and the detector casing 34 and the shell 42 are insulated by. In addition, 48 is an insulating material of glass or ceramics or the like for supporting and insulating the collector electrode 44.

[**0034**] In the gas introducing chamber 38, one side electrode 50 planar and parallel and close to the outer surface of α-ray incident window 36A are arranged while being insulated from other members. Further, while in parallel apart to the side electrode 50, a planar second side electrode 52 are disposed while being insulated from other members. On the other hand, between the side electrode 50 and the other side electrode 52, whereas the side electrode 50 high voltage is applied by the power supply 54 so that the negative electrode. For example, 1000~2000V is applied. On the other hand the distance between the side electrode 50 and the other side electrode 52 is 3cm for example. The second detection unit 28, except radon extraction filter 32 as described above, have the same configuration as the first detection unit, its description with reference numerals which are the number 100 in the same configuration as the first detection unit omitted.

[**0035**] Next, the operation of the two detection units.

[**0036**] If you want to monitor the radon-Tron concentration in the room, to place the two detection unit in the room. Then, as apparent from the above description, radon daughters and thoron daughters as a solid is blocked by radon extraction filter 32, also, becomes solid decay relatively short thoron half-life, radon extraction It is blocked by the filter 32. Therefore, in the first detection unit 26, only the radon daughter nuclides produced by decay after radon and entry exists. Radon daughter nuclides generated by the collapse shows the presence of radon as a parent nuclide was originally present. On the other hand, the second detection unit 28, since such a filter is not disposed, so that all the radon and its daughters and thoron and their daughter enters.

[0037] In the first detection unit, whereas a high voltage is applied electrical field between the negative electrode 50 and the other side electrode 52 is formed, however since the negative electrode 50 is a negative electrode, radon which is charged to the cathode daughter nuclides will adhere is attracted to one side electrode 50.

[0038] That is, since radon daughters is indicative entry was radon itself, and it is set in the vicinity of α-ray incident window 36A, is intended to increase the radon detection efficiency. In other words, while the α-rays generated by the attached daughter nuclide to the side electrode 50, enters passes through the α-ray incident window 36A to the ionization chamber 36, it is to ionize the internal gas, out of that time resulting ionization charge electronic components are collected in the collector electrode 44, is output to the outside as a pulse. Of course, alpha rays emitted from radon present in the gas introduction chamber 38 via the alpha ray incident window 36A enters into the ionization chamber 36, is detected.

[0039] Therefore, the ionization chamber 36 of the first detection unit 26 can detect the radon which has passed through the radon extraction filter 32, in which case, by the pair of electrodes 50 and 52, it is possible to increase the detection efficiency. The ratio of its daughter nuclides and parent nuclide in the equilibrium state is constant, if the detection efficiency of the parent nuclide and a detection efficiency of the daughter nuclide is determined by experiment or the like, the amount of parent nuclide from the detection results can be calculated easily .

[0040] On the other hand, in the second detection unit 28, it will be radon and its daughters and thoron and their daughter is introduced into the gas introducing chamber 138, similarly to the first detection unit 26, a charging nuclide radon daughter and thoron daughters would be trapped on the one side electrode 150. Then, α-rays emitted from radon progeny and thoron daughters, enters through the α-ray incident window 136A to the ionization chamber 136, is detected. Similarly, radon and α-rays emitted from the synchrotron in the gas introduction chamber 138 is also detected by the ionization chamber 136. Therefore, in the second detection 28 units, it is possible to obtain a detection value of the total. In that case, it is possible to perform the detection of α-ray summoned the daughter nuclide in the vicinity of α-ray incident window 136A, enhanced the detection sensitivity.

[0041] Incidentally, even if temporarily enters such other ions in the gas introduction chamber 138, because they are not detected unless radioactive ion pre recipe theta 12 shown in the apparatus of FIG. 4 is not required.

[0042] Returning to FIG. 1, it will be described the signal processing unit 30. Detection pulse 100 outputted from the first detection unit 26 is amplified by a preamplifier 56 and amplifier 58, passes through the waveform shaping circuit and the pulse height discriminator circuit, not shown, is input

to the counting circuit 60. Other counter circuit 60 in the pulse count is made, it is also executed efficiency correction, etc. with respect to the counted value. The output of the counting circuit 60 is as shown a count of radon alone.

[**0043**] On the other hand, the detection pulse output from the second detection unit 28 is amplified by a preamplifier 62 and amplifier 64, passes through the waveform shaping circuit and the pulse height discriminator circuit, not shown, is input to the counting circuit 66. Counting circuit counts the pulses, if necessary by performing the correction of the count value, and outputs the count value. Its output will exhibit all the count value of radon and its daughters and thoron and their daughter nuclides. According to this embodiment, it is also possible to determine the dose of the whole by the radon-series and thoron-series on the human body.

[**0044**] Known delay coincidence circuit 68 for receiving an output pulse of the second detection unit 28 is a circuit for obtaining a count of only Tron. Half-life of 216Po resulting from Tron is very short as 0.15 seconds, so that the continuous detection pulse of immediately 216Po subsequent to the detection pulse of this for Tron. Therefore, those utilizing the phenomenon, a certain period of time (e.g., one second) detecting when two pulses are continuous in the (detection of the continuous pulse), to estimate the detection Tron if it is detected it is. Of course, the probabilistic TRON - Two pulses outside 216Po can be detected continuous, the concentration of such radon monitor environmental radiation its probability for low low, less influence on measurement accuracy. The delay coincidence circuit 68, another count of said sequential pulses, to correct the count value if necessary.

[**0045**] Therefore, the output of the delay coincidence circuit 68 will exhibit a count value of Tron, given the output of the counter circuit 60 in the adding circuit 70 weights the number of pulses and the output of the predetermined delay coincidence circuit 68 Been by adding the number of weighting was performed pulses, it is possible to obtain the count value excluding the daughter nuclide (radon count + TRON count). In addition, in the difference arithmetic circuit 72, from the output of the counting circuit 66, by pulling the output of the output and the delayed coincidence counting circuit 68 of the counting circuit 60, a total of daughter nuclides, excluding the parent nuclide (radon daughters + Tron daughter nuclide) numerical value can be obtained.

[**0046**] In Figure 3, whereas specific examples of the side electrode 50 are shown. One side electrode 50, (A), the as shown in can be formed extremely thin conductive film shape, or (Bs), can also be a parallel wire structures. Or, it may be a grating wire structures as shown in (C). Incidentally, the other-side electrode 52 may employ various structures Similarly considering the inflow of the measurement gas.

[**0047**] In this embodiment, one side electrode 50, because the radionuclide attached thereto to accumulate increases with time, are interchangeable. Further, radon extraction filter 32 is also set to be replaceable.

[**0048**] As described above, according to this embodiment, the radon thoron can be monitored in total including their daughter nuclides, also can be obtained the breakdown. In particular, there is an advantage that the radon and thoron can be separated measurement. Then, it is possible to increase the detection sensitivity by the electric field formation can be realized detection of high accuracy.

[**0049**] As described above, according to the present invention, according to the present invention, the radon and thoron can each measured separately. In addition, it is possible to realize a measurement of radon progeny and thoron daughter nuclides. Furthermore, according to the present invention, the measurement sensitivity is improved, and can be easily economically measured in real time.

[**0050**] Brief description of the drawings

[**0051**] Is a block diagram showing an overall configuration of a radiation measuring device according to FIG 1 the present invention.

[**0052**] It is a diagram showing a concrete configuration of the detection unit 14 shown in FIG. 2 FIG.

[**0053**] Is a diagram illustrating an example of the structure of FIG. 3 electrodes.

[**0054**] Figure 4 is a diagram showing a configuration of a conventional radon thoron monitor.

[**0055**] It is a diagram illustrating a FIG. 5 radon-series and thoron-series.

[**0056**] Description of the code

[**0057**] 24 detection unit 26 first detection unit 28 and the second detection unit 30 signal processing unit 32 radon extraction filter 36, 136 ionization chamber 36A, 136A α-ray incident window 38, 138 gas introduction chamber 50, 150 one-side electrode 52, 152 the other side electrode

At least one radiation detection unit, a signal processing unit for treating a detection signal from the radiation detection unit, a radiation measuring device including a said radiation detecting unit includes a detector having a radiation entrance window, the radiation with the one side electrode disposed in

proximity to the entrance window, and the other side electrode disposed to face the one side electrode being spaced from said radiation incident window, a polarity to collect charging nuclide to said one side electrode, wherein be carried out and means for applying a voltage, the contrast detection of radiation by said detector by introducing a measurement gas between the the side electrode the other side electrodes during the other hand the side electrode and the other side electrode radiation measurement apparatus according to claim.

Processing a first detection unit for detecting the radon, and a second detection unit for detecting the radon and its daughters and thoron and their daughter nuclide, a detection signal outputted from said first and second detection units a radon thoron measuring apparatus including a signal processing unit, and the first detection unit has a first and ionization chamber having an α-ray incident window, the first ionization across said α-ray incident window a first gas introducing chamber formed next to the box, the disposed first gas inlet of the gas introduction chamber, a filter for transmitting radon gas only, to the radiation entrance window in the first gas introducing chamber a first one side electrode disposed in proximity to a first other-side electrode disposed opposite to the one side electrode being spaced from said radiation incident window in the first gas introducing chamber, wherein the radon daughter nuclides generated by the decay after filter transmission has a polarity to aggregate the one side electrode, a first voltage applying means for applying a voltage between said one side electrode and the other side electrode, said second detection unit, a second ionization chamber having an α-ray incident window, the formed next to the second ionization chamber with respect to the α-ray incident window, from the gas inlet a second that the measurement gas is introduced a gas introducing chamber, and a second one side electrode disposed in proximity to the α-ray incident window in the second gas introducing chamber, while being separated from the α-ray incident window in the second gas introducing chamber wherein a second other-side electrode disposed opposite to the second one-side electrodes, with a polarity to collect the radon daughters and thoron daughters in the second one side electrode, the second one side electrode radon thoron measuring apparatus characterized by comprising a second voltage applying means for applying a voltage, the between the second other-side electrode and the.

The apparatus of claim 2, wherein said signal processing unit, radon thoron measuring apparatus which comprises means for determining radon count, means for determining Tron counts, the.